

1
2 Personalized AI assistants could either promote or hinder the
3 emergence of collective intelligence [SWP: *The potential*
4 *impact of personalized AI assistants on collective*
5 *intelligence?*]

6
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11 Abstract

12 Recent advances in artificial intelligence (AI) have enabled widespread use of per-
13 sonal AI assistants for various cognitive tasks, ranging from simple everyday tasks
14 to scientific discovery and creative writing. While AI assistance can improve the
15 performance of each individual, it can weaken their subsequent performance with-
16 out AI and further limit the diversity of interests and opinions. These risks raise
17 the question of whether human collective intelligence can persist as our society be-
18 comes increasingly reliant on AI assistants. To address this question, we develop an
19 evolutionary model of collective intelligence, where each individual participate in a
20 prediction task, consult AI, and imitate each other’s prediction strategy. We find
21 that “Chatbot” AI, which depends on users’ interests, can prevent the emergence
22 of collective intelligence by either limiting diversity or propagating systematic bias
23 through overreliance. Directly incentivizing or penalizing AI use generally worsens
24 these outcomes. Instead, a balanced incentive structure that preserve minority ex-
25 pertise and reward contributions to collective outcome can promote the emergence of
26 collective intelligence. However, the inherent ability of the population’s to make pre-
27 diction in the absence of AI remains suboptimal even under this balanced incentive
28 structure, raising concerns for continued AI reliance. [SWP: *conclusion?*]

29 Introduction

30 Collective intelligence (CI) describes a phenomenon in which a group’s ability to
31 solve problems, make decisions, or adapt to changing environments exceeds each
32 member’s ability (Galton, 1907; Woolley et al., 2010; Yaniv and Milyavsky, 2007;
33 Hommes et al., 2005; Lorge et al., 1958). This phenomenon, also referred to as
34 wisdom of crowds, has thus served as a central mechanism underlying the success of
35 diverse organizations, ranging from animal groups (Dreyer et al., 2025; Simons, 2004;
36 Berdahl et al., 2013) and human societies (Woolley et al., 2010; Riedl et al., 2021) to
37 machine-learning systems (Breiman, 1996, 2001). Understanding the conditions that
38 promote and maintain CI therefore remains a fundamental question across social
39 sciences (Mann and Helbing, 2017), evolutionary biology (Wang et al., 2025), and
40 computer science (Werfel, 2026). Addressing this question has become particularly
41 timely as artificial intelligence (AI) has begun to reshape how individuals acquire
42 information, make decisions, and contribute to collective outcomes (Cui and Yasseri,
43 2024).

44 The recent emergence of generative AIs and large language models (LLMs) has
45 changed the way individuals engage in various cognitive tasks, ranging from simple
46 everyday tasks to more complex tasks (Noy and Zhang, 2023; Brynjolfsson et al.,
47 2025), including scientific discovery (Lu et al., 2026) and creative writing (Doshi
48 and Hauser, 2024). In particular, while the use of generative AI models, such as
49 ChatGPT, has been rapidly increasing (Bick et al., 2026; Sajadieh et al., 2026), the
50 potential long-term impact of personal AI assistants on human society, especially the
51 CI, remains uncertain. Some argue that AI can promote CI by enhancing individual
52 expertise (Vaccaro et al., 2024), and opinion aggregation (Burton et al., 2024; Tessler
53 et al., 2024). Conversely, empirical evidence indicate that the use of AIs can limit the
54 diversity of opinions (Fügener et al., 2021; Doshi and Hauser, 2024; Lee and Chung,
55 2024; Meincke et al., 2025), a key requirement for the emergence of CI (Hong and
56 Page, 2004; Lorenz et al., 2011; Aplin et al., 2014; Becker et al., 2017).

57 Our ability to predict the long-term impact of personal AI assistants on human
58 CI is further hampered by the apparent discrepancy in the incentives that shape AI
59 uses. On one hand, many industrial sectors are pushing towards increased AI usages,
60 where the number of “tokens” used is often perceived as a proxy for productivity
61 (Bousquette, 2026; Keary, 2026). On the other hand, many academic institutions
62 appear to discourage, and even penalize, AI uses, reflecting concerns about plagia-
63 rism, misinformation, and limited independence (Thorp, 2023; Chawla, 2026). Such
64 discrepancy raises a question about whether directly incentivizing or penalizing AI
65 usage will benefit or harm human CI. These observations further highlight the im-
66 portance of indentifying a robust incentive structure that can promote CI in the
67 presence of personal AI assistants. Such incentive structure should seek to retain
68 the benefits of AIs and prevent potential risk, such as preserving the diversity of
69 individual opinions.

70 To address these questions, we extend previous approaches for modeling CI, where

each individual participates in a prediction task from partial observations of underlying causal factors (Mann and Helbing, 2017; Wang et al., 2025; Hong et al., 2012). In particular, we allow each individual to independently consult with personal AI assistants and revise their predictions accordingly. The resulting model allows us to investigate the robustness of human CIs under different assumptions about the AI model’s ability to make predictions as well as incentive structures that are associated with the prediction task. In doing so, we assess the impact of directly incentivizing or penalizing AI uses on CI and further suggest a balanced incentive structure that can maintain CI in the presence of AI.

Mathematical model

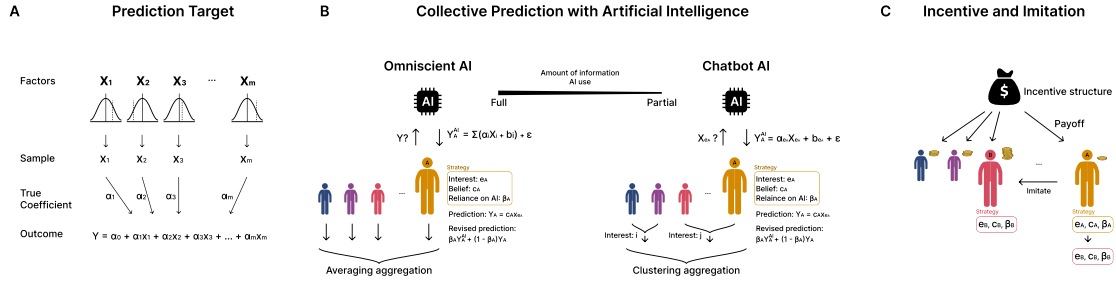


Figure 1: **A schematic diagram of the model for individual and collective prediction in the presence of personal AI assistants.** (A) The prediction target is modeled as a linear combination of causal factors. (B) Each player is characterized by three strategies: interest, belief, and reliance on AI. Players make predictions based on their interests and beliefs and subsequently revise them with AI assistance. We consider two types of AI, Chatbot AI and Omniscient AI, which differ in the amount of information available for generating predictions. (C) Players receive payoffs according to the incentive structure, and the population evolves through imitation of peers with higher payoffs.

We begin by extending previously established models of CI, in which a group of players participates in a prediction task (Figure 1): each player makes an individual prediction based on partial observations of the underlying causal factors, and the group forms a collective prediction by aggregating these individual predictions (Hong et al., 2012; Mann and Helbing, 2017; Wang et al., 2025). Specifically, the prediction target, Y , is modeled as a linear combination of M causal factors x_i , with fixed coefficients α_i : $Y = \sum_{i=1}^M \alpha_i x_i + \alpha_0$, where each factor is drawn from independent normal distribution (Figure 1A). Then, each individual A observes a single factor of their interest x_{e_A} and makes a prediction based on their own belief, c_A : $Y_A = c_A x_{e_A}$ (Figure 1B). We assume each individual can then independently consult their AI assistants and revise their prediction based on their reliance on AI, β_A :

92 $Y_A^{\text{re}} = \beta_A Y_A^{\text{AI}} + (1 - \beta_A) Y_A$ (Figure 1B). The revised predictions are then aggregated
 93 using two alternative aggregation rules, averaging and clustering, to form collective
 94 prediction.

95 To model AI, we assume that AI has already been trained on similar tasks and is
 96 able to make predictions based on partial or full observations of causal factors (Figure
 97 1B). Every time a player consults AI, it will make a new prediction Y_A^{AI} based on
 98 its internal parameters, which can include factor-specific bias (b_i) and stochastic
 99 error ϵ . Here, we consider two extreme types of AI response, which differ in the
 100 amount of information AI uses when making predictions. In one extreme, the AI
 101 can observe all causal factors and use all information to make predictions, regardless
 102 of the user’s interest: $Y_A^{\text{AI}} = \sum_{i=0}^M (\alpha_i x_i + b_i) + \epsilon$. In the other extreme, AI works
 103 like a chatbot, where the AI predicts the outcome based only on the user’s interest:
 104 $Y_A^{\text{AI}} = \alpha_{e_A} x_{e_A} + b_{e_A} + \epsilon$. Hereafter, we refer to these two types as ‘Omniscient AI’
 105 and ‘Chatbot AI’, respectively.

106 We assume that appropriate aggregation rule for collective prediction depends
 107 on the type of AI. When Chatbot AI is used, each revised prediction depends on a
 108 single factor. Thus, we use clustering aggregation, under which revised predictions
 109 are first averaged among individuals who are interested in the same factor, and
 110 then summed across factors. When Omniscient AI is used, however, each revised
 111 prediction contains information about all causal factors, in which case clustering
 112 these predictions by interest would be inappropriate. Instead, we use averaging
 113 aggregation, under which revised individual predictions are averaged directly to form
 114 the collective prediction.

115 Each generation consists of multiple round of predictions, and each player is re-
 116 warded a payoff according to the incentive structure (Figure 1C). For simplicity,
 117 we assume that a sufficiently large number of prediction rounds occurs within each
 118 generation, allowing us to use the expected payoff value rather than computing the
 119 payoffs for every round. We consider two incentive structures from previous work,
 120 “feedback” and “niche-expert”, which have been shown to promote near-perfect col-
 121 lective accuracy in the absence of AI (Wang et al., 2025). The feedback structure
 122 gives higher payoff to individuals who can push the collective prediction toward the
 123 true target, regardless of their individual predictive accuracy: $\pi_A = Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}})$. In
 124 contrast, the niche-expert structure gives higher payoff to individuals who have both
 125 better predictive accuracy and a minority interest: $\pi_A = -\rho_{e_A}(Y_A^{\text{re}} - Y)^2$. Here, ρ_{e_A}
 126 denotes the proportion of players interested in the same factor, thereby allowing the
 127 niche-expert structure to preserve interest diversity. We note collective intelligence
 128 fails to emerge under the combination of averaging aggregation with niche-expert
 129 structure in the absence of AI (Wang et al., 2025).

130 Through time, the group evolves by allowing each individual to imitate peers
 131 who received higher payoffs (Figure 1C). Specifically, after each generation, two
 132 individuals are randomly sampled, and one imitates all three strategies (the factor of
 133 interest, factor belief, and reliance on AI) of the other with a probability determined
 134 by the difference in their expected payoffs (Materials and Methods). Individual

135 predictions are aggregated into a collective prediction $\hat{Y}$ based on the aggregation
 136 rules. Collective accuracy is defined as the reduction in prediction error achieved
 137 by the collective prediction relative to a baseline prediction α_0 (Wang et al., 2025):
 138 $1 - \mathbb{E}[(Y - \hat{Y}^{\text{re}})^2] / \mathbb{E}[(Y - \alpha_0)^2]$. Hereafter, we say that CI emerges when the collective
 139 accuracy reaches a near perfect level.

140 This model now allows us to explore the impact of AI on CI depending on the
 141 mode of AI and the incentive structure. Here, we begin by comparing the emergence
 142 of CI with and without AI and subsequently investigate a how different incentive
 143 structures affect the emergence of CI in the presence of AI.

144 AI can either promote or prevent the emergence of 145 CI

146 To understand the impact of AI, we first compare collective dynamics with and
 147 without AI. First, consistent with Wang et al. (2025), simple imitation rules and
 148 incentive structures allow collective intelligence to emerge in the absence of AI (Fig.
 149 2A,C,D). One exception is the use of simple averaging with niche-expert incentive
 150 structures, under which the collective accuracy of the population remains near zero.
 151 *[JS: This is because enhancing individual accuracy causes individuals to predict*
 152 *close to the true coefficient of their interest, but simply averaging these predictions*
 153 *does not necessarily produce an accurate estimate of the full prediction target.]* (Fig.
 154 2B).

155 We find that AI usage can have heterogeneous outcomes depending on its type.
 156 For example, the use of Omniscient AI can either preserve CI under the feedback
 157 incentive structure (Fig. 2A) or promote higher collective accuracy under the niche-
 158 expert incentive structure (Fig. 2B). These results are robust across different levels
 159 of AI accuracy and can be shown analytically by assuming a sufficiently large group
 160 of players (Fig. S1; see SI Evolutionary dynamics section). *[SWP: need to get*
 161 *back.]*

162 In contrast, the use of Chatbot AI prevents the emergence of CI, thereby resulting
 163 in lower levels of collective accuracy (Fig. 2C, D). Under the feedback incentive
 164 structure, the use of Chatbot AI leads to a loss of diversity in the population's
 165 interest, making it impossible to account for effect of some causal factors (Fig. S1,
 166 S2). This causes an increase in collective variance and stochastic decrease in collective
 167 accuracy (Fig. 1; Fig. S1). We note that the loss of interest diversity may arise even
 168 in the absence of AI in finite populations (Fig. S3), but the use of Chatbot AI
 169 makes the group more vulnerable to such loss (Fig. S3, S4). Under the niche-
 170 expert incentive structure, the population becomes overly reliant on AI, causing the
 171 collective prediction to inherit the AI's bias (Fig. S2). This leads to an increase in
 172 collective bias and a decrease in collective accuracy (Fig. 2; Fig. S2). Interestingly,
 173 we find that the niche-expert structure can prevent the use of AI under specific
 174 assumptions about the relationship between the true coefficient and AI bias (Fig.

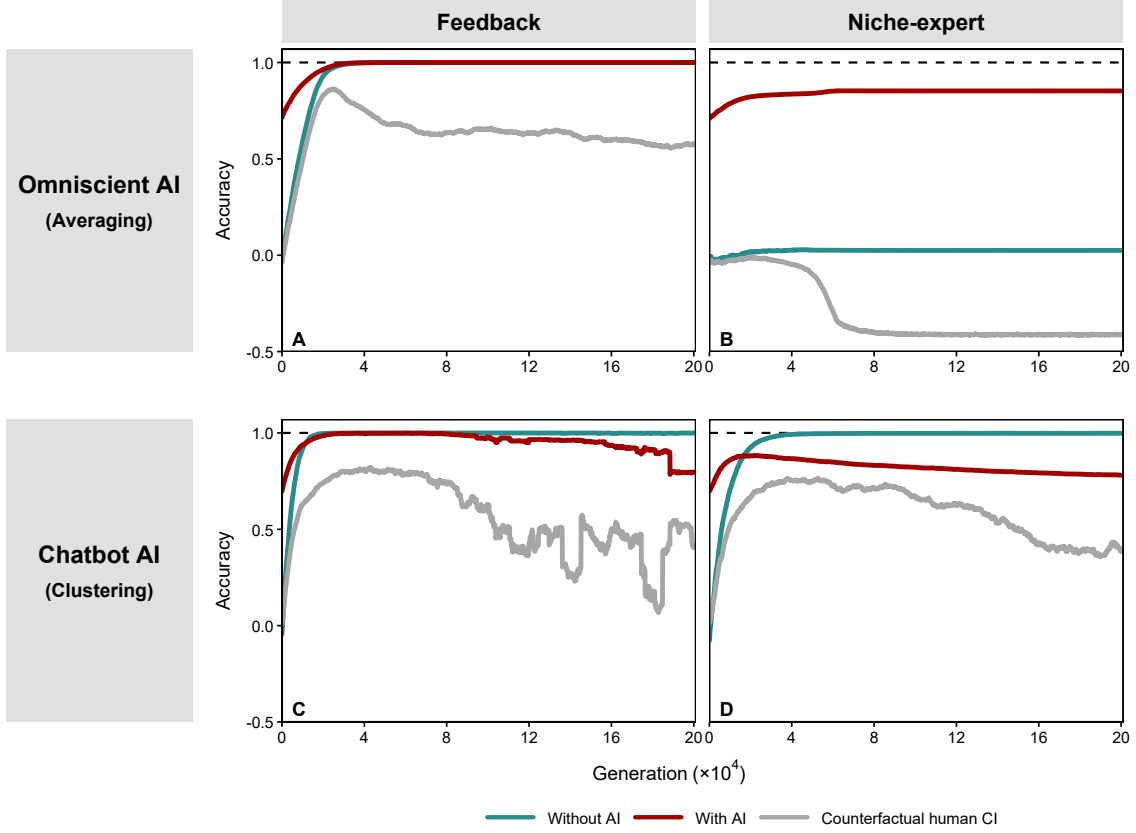


Figure 2: **Comparison of collective accuracy with and without AI.** Trajectories of collective accuracy when the group is assisted by the Omniscient AI (A,B) and Chatbot AI (C,D). Here, feedback (A,C) and niche-expert (B,D) incentive structures are considered. Green lines represent trajectories of collective accuracy without AI assistance. Red lines represent trajectories of collective accuracy with AI assistance. Gray lines represent trajectories of counterfactual collective accuracy, computed from players’ unrevised predictions, indicating the group’s predictive accuracy if AI assistance were unavailable at the given generation. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 100^2)$ for (A, B), $c_A \sim \mathcal{N}(0, 5^2)$ for (C, D), and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.36$ and $\tau = 0.3$.

175 S1). In this case, the population completely excludes AI assistance, allowing the
 176 CI to emerge eventually. However, this phenomenon can substantially delay the
 177 emergence of CI (Fig. S5).

178 To investigate whether the group can retain its predictive ability when AI assis-
 179 tance suddenly becomes unavailable at any given time, we quantify counterfactual
 180 collective accuracy based on unrevised predictions. Across all scenarios, we find that
 181 the counterfactual collective accuracy is lower than the collective accuracy of the

182 corresponding population.

183 In summary, an omniscient AI can promote CI, but AI operating with limited in-
 184 formation may prevent the emergence CI. Moreover, reliance on AI can limit inherent
 185 ability of the population to make accurate predictions on their own, raising concerns
 186 about future AI usage. These findings further underscore the need for incentive
 187 structures that can mitigate the potential risks associated with AI assistance.

188 Directly incentivizing or penalizing AI use can ad- 189 ditionally harm CI

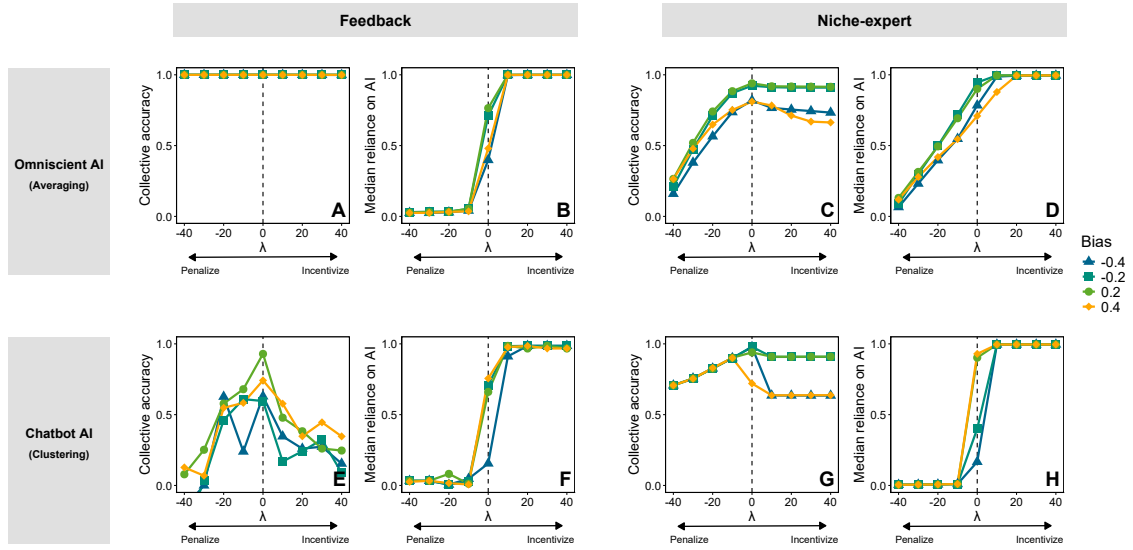


Figure 3: **Collective accuracy under the influence of direct intervention on AI use.** Collective accuracy and median reliance on AI under varying strengths of intervention (λ) across different assumptions about AI types (omniscient AI, A–D and chatbot AI, E–H) and incentive structures (Feedback, A,B,E,F, and Niche expert, C,D,G,H). Positive λ represents an incentive for AI use, whereas negative λ represents a penalty for AI use. In each panel, the vertical dashed line indicates the original incentive structure ($\lambda = 0$). Four AI bias levels are shown: $b_i = \pm 0.2$, representing higher-accuracy AI, and $b_i = \pm 0.4$, representing lower-accuracy AI. Collective accuracy (A,C,E,G) and median reliance on AI (B,D,F,H) are averaged over generations 190,000–200,000. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 100^2)$ for (A–D), $c_A \sim \mathcal{N}(0, 5^2)$ for (E–H), and $\beta_A \sim U(0, 1)$. AI parameter: $\tau = 0.3$.

190 Given the strong influence of AI on collective outcomes, a straightforward inter-
 191 vention is to directly incentivize or penalize the use of AI. To investigate whether

such direct incentives or penalties can help the emergence of CI, we modify the two incentive structures by adding a payoff term proportional to individual reliance on AI:

$$\text{Feedback/Niche-expert} + \lambda\beta_A \quad (1)$$

where λ represent the strength of intervention ($\lambda > 0$) or penalty ($\lambda < 0$). [SWP: maybe we need to call incentive pi earlier on...] [SWP: also good to define beta again here as a reminder.]

Directly incentivizing or penalizing AI use can further reduce the collective accuracy (Fig. 3C, E, G), except when Omniscient AI is used under the feedback structure; in the latter case, the population retains collective intelligence regardless of the intervention strength (Fig. 3A). The reduction in collective accuracy arises because incentivizing or penalizing AI usage causes individuals to imitate others based on their AI usage rather than their predictive accuracy. Thus, stronger interventions can push the group to always ($\beta = 1$) or never ($\beta = 0$) rely on AI (Fig. 3B,D, F, H) but these scenarios do not necessarily lead to better outcomes (Fig. 3D, F, H). When AI bias is relatively large, a weak penalty can lead to higher collective accuracy, but this represents a narrow parameter regime (Figure 3G).

Overall, these results indicate that simply incentivizing or penalizing the AI, independent of individuals' performance, use may exacerbate rather than resolve the risk of AI on CI. Thus, we also considered an incentive structure where interventions for AI usage affects the payoffs multiplicatively, thus giving higher payoffs to individuals who rely more (or less) AI and makes better prediction. [SWP: Supp.]

Balanced incentives mitigate AI-induced failures of CI

Neither the original incentive structures nor simple interventions targeting AI use are sufficient to promote CI. What incentive structure, then, would work in the presence of AI? Here, we propose a “balanced” incentive structure, modeled as the average of the feedback and niche expert incentives.

The core idea behind the balanced structure is to maintain interest diversity and prevent collective bias. On one hand, the feedback incentive structure is prone to the loss of interest diversity but can prevent collective bias by incentivizing players whose predictions lie opposite to the collective prediction. On the other hand, the niche-expert structure can maintain interest diversity because it directly incentivizes minority interests. However the niche-expert structure cannot prevent collective bias because, within the same factor of interest, individuals are rewarded only according to their individual predictive performance, which in turn homogenize the predictions.

Therefore, when the two incentive structures are combined, the population achieves CI by maintaining interest diversity while limiting collective bias (Fig. 4A; Fig. S3). Under the balanced structure, individuals remain more evenly distributed across causal factors, and their beliefs lie closer to the true coefficients than under either

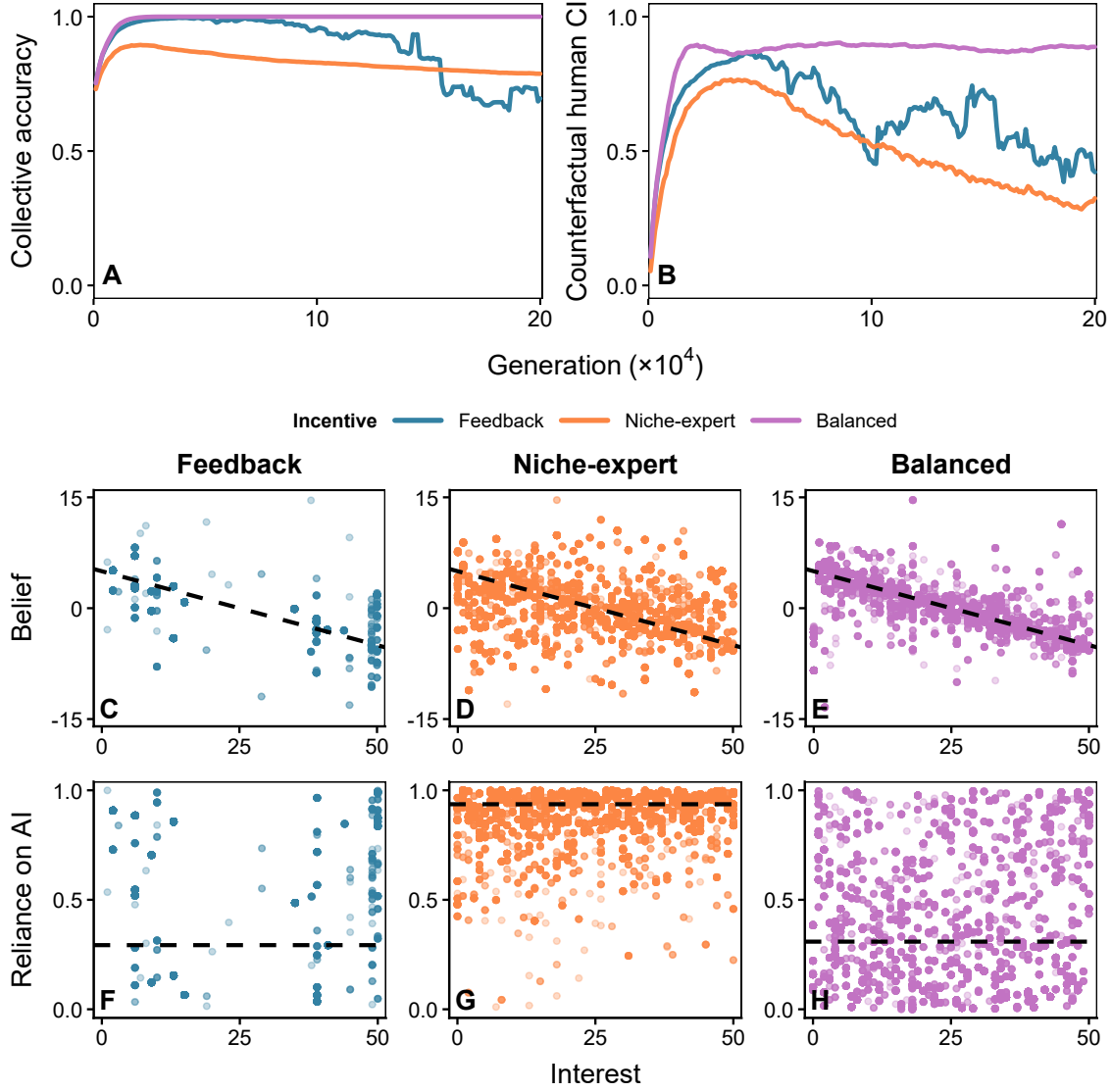


Figure 4: **Balanced structure can promote collective intelligence** Trajectories of collective accuracy (A) and counterfactual human CI (B) under three incentive structures (feedback, niche-expert, and balanced) with Chatbot AI. *[SWP: A and B should have the same y axes scaling.]* Distributions of individual interest and belief (C-E) and of interest and reliance on AI (F-H) after 200,000 generations. Each point represents an individual. Dashed lines in panels C-E represent the true coefficients (α_i) of the prediction target. Dashed horizontal lines in panels F-H indicate the median reliance on AI. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i = 5 - 0.2i$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 5^2)$, and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.4$ and $\tau = 0.3$.

incentive structure alone (Fig. 4C–E). The balanced structure also produces higher counterfactual collective accuracy than the feedback or niche-expert structure (Fig. 4B). These patterns are driven by the emergent property of the balanced structure, under which the population increases its reliance on AI as AI becomes more accurate.

[SWP: *may need some robustness check rather than a case study...*]

Discussion

The ongoing race towards the development of “better” AIs continues to pose questions about the societal benefits and risks of AI uses (Bender et al., 2021; Burton et al., 2024; Cui and Yasseri, 2024). Here, we used mathematical models of evolutionary dynamics to investigate how AI uses and incentive structures jointly affect the emergence of CI, which is defined as the population’s ability to make accurate predictions. We find that an “omniscient” AI model, which incorporates information about all relevant causal factors, can improve collective accuracy, whereas a “chatbot” AI model, whose outcome depends on the user’s interest, can impair it. These contrasting outcomes arise because AI assistance affects not only individual predictions, but also the population’s interest diversity, beliefs, and reliance on AI. We also show that directly incentivizing or penalizing AI uses, independent of individual’s performances, can further limit the collective accuracy of the population. Instead, a balanced incentive structure can maintain CI by promoting diversity and limiting bias.

Our analysis reveals that AI can prevent the emergence of CI through two different mechanisms. First, AI use can cause individuals to concentrate on a subset of the factors that determine the prediction target, thereby reducing diversity in their interests. This result is broadly consistent with recent empirical studies showing that AI assistance can improve individual performance but limit the diversity of outputs, in domains ranging from scientific research (Hao et al., 2026) to creative writing (Doshi and Hauser, 2024). Second, over-reliance on AI can cause the population to inherit inherent biases in AI’s predictions (Fügener et al., 2021; Vicente and Matute, 2023), causing the prediction accuracy to decrease eventually. This mechanism can prevent the emergence of CI even when the interest diversity is maintained because biases in AI predictions can be amplified when individual predictions are aggregated. Thus, preserving the diversity of ideas will not be sufficient to promote CI. Instead, AI uses need to be calibrated to prevent systematic AI biases from becoming collective biases (Buçinca et al., 2021).

Our analyses of counterfactual collective accuracy also raises an important concern for future AI uses: while AI uses may improve the apparent performance of individuals and, in some cases, the population as a whole, it can simultaneously reduce the population’s ability to make accurate predictions without AI. These results echo findings from a recent field experiment, which demonstrated that students, who seemingly performed better in solving mathematics problems under AI assis-

271 tance, performed worse on their own than those who did not have access (Bastani
 272 et al., 2025). These findings highlight the need to evaluate AI tools not only by
 273 the performance gains they provide during use, but also by whether individuals and
 274 populations retain the ability to perform without them (Melumad and Yun, 2025;
 275 Ke et al., 2026). Making this distinction clear will be particularly important for
 276 education, where AI may help students complete a task without helping them learn
 277 how to do it on their own (Bastani et al., 2025; Melumad and Yun, 2025).

278 As generative AIs and large language models are already generating substantial
 279 benefits, many organizations, both industrial and academic, are increasingly seek-
 280 ing to improve their collective performance by regulating AI uses (Thorp, 2023;
 281 Bousquette, 2026; Chawla, 2026; Keary, 2026). Such regulations may take direct
 282 forms, including AI-use policies or token-based leaderboards, or may operate indi-
 283 rectly through other incentives, such as productivity rewards or the social stigma
 284 associated with AI use (Reif et al., 2025). Our model suggests that directly incen-
 285 tivizing or penalizing AI use is unlikely to induce appropriately calibrated reliance on
 286 AI or to promote CI. Incentivizing AI use could lead to overreliance, causing groups
 287 to propagate AI biases, whereas penalizing AI use could lead to underreliance, pre-
 288 venting groups from using AI even when it performs well in the relevant domain.
 289 Our result instead shows that appropriate AI use and accurate CI can be promoted
 290 by focusing on the traditional properties of groups that have been emphasized for
 291 decades: maintaining diversity and preventing collective bias, rather than on AI use
 292 itself (Hong and Page, 2004; Lorenz et al., 2011; Mann and Helbing, 2017).

293 There are several limitations to our analyses. First, our model allows for social
 294 learning among humans, but AI’s inherent ability remains fixed throughout the evo-
 295 lutionary process. In reality, AI systems can also learn from human-generated data
 296 (Ouyang et al., 2022). Such bidirectional learning increase the similarity between
 297 human and AI errors and therefore further reduce the group performance (?). Sec-
 298 ond, we model AI errors using fixed, factor-specific additive biases and independent
 299 random errors. This assumption allows random errors to be reduced through ag-
 300 gregation, while systematic biases can still persist in the collective prediction. In
 301 reality, AI errors may have more complex structures, including biased estimates of
 302 factor effects and correlated errors across individuals, factors, or repeated interac-
 303 tions. Further empirical and theoretical work is therefore needed to understand how
 304 these error structures affect the emergence of CI. Finally, following established mod-
 305 els of CI, we assume that the prediction target is a linear combination of independent
 306 causal factors. This assumption does not hold for many real-world tasks, where causal
 307 factors may be correlated or interact nonlinearly. Future studies should therefore in-
 308 vestigate how nonlinear relationships between causal factors and prediction targets
 309 affect the interaction between AI assistance and the emergence of CI.

310 The rapid adoption of personalized AI assistants is changing how individuals
 311 learn, work, and make decisions. While the immediate benefits and risks of these
 312 systems have been evaluated at the individual level, our study raises a broader con-
 313 cern that their widespread AI use may reshape collective intelligence of human so-

ciety. Our results further underline the importance of designing reward structures that preserve minority expertise and prevent AI biases from persisting across the population. **[SWP: need a concluding sentence!]**

Methods

Prediction target

Following previous models of CI, we model the prediction target as a linear combination of M independent causal factors (Hong et al., 2012; Mann and Helbing, 2017; Wang et al., 2025):

$$Y = \alpha_0 + \alpha_1 X_1 + \dots + \alpha_M X_M, \quad (2)$$

where the coefficient α_i determines the magnitude and direction of the effect of factor i , and α_0 represents a global intercept. Each causal factor X_i is modeled as an independent Gaussian random variable with mean zero and standard deviation σ_i .

Players

We consider a well-mixed population $\mathcal{N}$ consisting of N players engaged in a novel prediction task. In the absence of AI assistance, the strategy of player A is characterized by two variables: interest e_A and belief c_A . Player A observes the causal factor X_{e_A} and makes a prediction by multiplying the observed factor value by their belief:

$$Y_A = c_A X_{e_A}, \quad (3)$$

where e_A can take an integer value from 0 to M . When $e_A = 0$, the player does not observe any causal factor and instead makes a constant prediction based solely on their belief. We refer to this as a stubborn strategy because the player’s prediction does not respond to the realized values of the causal factors. For notational convenience, we assume that $X_0 = 1$.

When the group is assisted by AI, each player’s strategy additionally includes reliance on AI, denoted by β_A . We assume that each player consults the AI independently and revises their prediction by taking a weighted average of their own prediction and the AI-generated prediction:

$$Y_A^{\text{re}} = \beta_A Y_A^{\text{AI}} + (1 - \beta_A) Y_A, \quad (4)$$

where β_A ranges from 0 to 1. A value of $\beta_A = 1$ indicates complete reliance on AI, whereas $\beta_A = 0$ indicates that the player does not use the AI prediction.

AI models

We assume that the AI model has been trained on similar tasks and is therefore capable of making predictions based on causal factors. However, because the model

is imperfect, its outputs are subject to systematic additive bias and stochastic error. The additive bias associated with factor i is denoted by b_i , and the stochastic error is denoted by ϵ . The AI is neither trained nor updated during the simulation, and its biases therefore remain fixed throughout the simulation. For simplicity, we assume $b_i = b$ for the main analysis for all i . The model formulation nevertheless allows biases to differ among factors.

We assume that the stochastic error follows a Gaussian distribution with mean zero and standard deviation τ . Each time a player consults the AI, an independent error is generated. Errors are therefore independent across players and across repeated consultations by the same player. Consequently, stochastic AI errors can be attenuated when independently generated predictions are aggregated across players. We therefore hold τ constant and vary the magnitude of the additive bias to manipulate the level of AI accuracy during our analyses.

We consider two extreme types of AI that differ in the amount of information available for prediction. The first is termed the Omniscient AI, which can observe all M causal factors and incorporate them into its prediction regardless of the user's interest. The prediction generated by the Omniscient AI for player A is expressed as follows:

$$Y_A^{\text{AI}} = \alpha_0 + b_0 + \alpha_1 X_1 + b_1 + \cdots + \alpha_M X_M + b_M + \epsilon. \quad (5)$$

The second is termed the Chatbot AI, which can use only the single causal factor X_{e_A} associated with the player's interest. The prediction generated by the Chatbot AI for player A is expressed as follows:

$$Y_A^{\text{AI}} = \alpha_{e_A} X_{e_A} + b_{e_A} + \epsilon. \quad (6)$$

Aggregation rules

The revised individual predictions are aggregated to form a collective prediction. We use two aggregation rules proposed in previous research (Wang et al., 2025), depending on the type of AI used by the group. First, when the group is assisted by the Omniscient AI, the AI component of each revised prediction contains information about all M factors, even though the individual's own prediction is based on only one factor. Each revised prediction can therefore be treated as an estimate of the entire target. Accordingly, we calculate the collective prediction by taking the simple average of the revised individual predictions:

$$\hat{Y}^{\text{re}} = \frac{Y_1^{\text{re}} + Y_2^{\text{re}} + \cdots + Y_n^{\text{re}}}{N}. \quad (7)$$

Second, when the group is assisted by the Chatbot AI, both the player's initial prediction and the AI's prediction are based on the same causal factor. Each revised prediction therefore contains information about only one factor and can be treated as an estimate of that factor's contribution to the target. We consequently use clustering aggregation.

380 Clustering aggregation is performed by grouping individuals according to their
 381 interests and calculating the mean prediction within each group. The collective
 382 prediction is then obtained by summing the mean predictions of all nonempty groups:

$$\hat{Y}^{\text{re}} = \sum_{i:n_i>0} \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} Y_A^{\text{re}}, \quad (8)$$

383 where $\mathcal{A}_i$ represents the set of players interested in factor i , and $n_i = |\mathcal{A}_i|$ represents
 384 the number of players in that set.

385 When the group is not assisted by AI, collective predictions are computed from the
 386 unrevised individual predictions throughout the simulation. Both simple averaging
 387 and clustering can be used in the absence of AI. For fair comparison, we use the
 388 same aggregation rule as in the corresponding AI-assisted condition.

389 Incentive structures

390 Players receive payoffs based on the underlying incentive structure. We consider five
 391 incentive structures: feedback structure, niche-expert structure, direct intervention
 392 on AI use of these two structures, and a balanced structure.

393 The feedback structure does not directly reward individuals according to their
 394 own prediction accuracy. Instead, it rewards predictions that are aligned with the
 395 residual error of the collective prediction and therefore have the potential to move
 396 the collective prediction toward the true value. Given player A 's revised prediction
 397 Y_A^{re} and the collective prediction $\hat{Y}^{\text{re}}$, the payoff is expressed as

$$\pi_A = Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}). \quad (9)$$

398 The niche-expert structure assigns higher payoffs to individuals who make accu-
 399 rate predictions while having an underrepresented interest. Specifically, the squared
 400 prediction error of player A is weighted by $\rho_{e_A} = n_{e_A}/n$, the proportion of players
 401 interested in the same factor:

$$\pi_A = -\rho_{e_A}(Y_A^{\text{re}} - Y)^2. \quad (10)$$

402 Previous research has shown that these two incentive structures can produce near
 403 perfect collective accuracy through social learning, except when the niche-expert
 404 incentive is combined with simple averaging (Wang et al., 2025). They are therefore
 405 used as baseline structures for examining how AI assistance affects the emergence of
 406 CI.

407 To examine the effects of directly encouraging or discouraging AI use, we mod-
 408 ify the feedback and niche-expert structures by adding a term proportional to the
 409 player's reliance on AI:

$$\pi_A = Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) + \lambda\beta_A \quad (11)$$

$$\pi_A = -\rho_{e_A}(Y_A^{\text{re}} - Y)^2 + \lambda\beta_A. \quad (12)$$

Here, λ represents the strength of the direct incentive or penalty applied to AI use. A positive value of λ rewards AI reliance, whereas a negative value penalizes it. When $\lambda = 0$, no direct incentive or penalty is applied, and the payoff functions reduce to the original feedback and niche-expert structures.

The balanced structure is defined as the arithmetic mean of the feedback and niche-expert payoff components:

$$\pi_A = \frac{Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) - \rho_{e_A}(Y_A^{\text{re}} - Y)^2}{2}. \quad (13)$$

Simulations

All simulations are conducted using an agent-based model with 50 causal factors ($M = 50$) and 10,000 players ($n = 10,000$). The parameters defining the prediction target are sampled at the beginning of the simulations and subsequently held fixed.

Coefficients α_i are sampled from a uniform distribution on $[-5, 5]$, except in the simulation presented in Figure 4, in which α_i is set to $5 - 0.2i$ for visualization. Standard deviations σ_i are sampled from a uniform distribution on $[0, 3]$.

Players' initial strategies are also sampled from predefined probability distributions. The distribution of individual belief c_A depends on the aggregation rule. Under simple averaging, each factor's contribution is scaled by the proportion of players interested in that factor. We therefore sample beliefs from a Gaussian distribution with mean zero and standard deviation 100, following the scaling used in previous work (Wang et al., 2025). Under clustering aggregation, beliefs are sampled from a Gaussian distribution with mean zero and standard deviation 5. Because predictions are averaged separately within each interest group and then summed, no additional scaling of c_A is required. The initial interest of each player is sampled uniformly from the integers between 0 and 50, and reliance on AI was sampled from a uniform distribution on $[0, 1]$.

We assume that prediction events occur on a substantially faster timescale than social learning. In each generation, we therefore calculate the expected payoff $\mathbb{E}[\pi_A]$ of each player analytically rather than explicitly sampling prediction events. The expectation is taken over the distributions of the causal factors $X_1, \dots, X_M$ and the stochastic AI error ϵ , conditional on the current population strategies and the fixed target parameters. Closed form expression of expected payoffs are derived for all incentive structures and are presented in the Supplementary Methods.

Each generation consists of an expected payoff calculation followed by one opportunity for pairwise social learning. Two distinct players, A and B , are randomly sampled, and player A imitated all three strategies of player B with probability

$$\frac{1}{1 + \exp(s(\mathbb{E}[\pi_A] - \mathbb{E}[\pi_B]))}, \quad (14)$$

where s represents the strength of social learning and was set to 50 in all simulations. We introduce a mutation rate μ , such that with probability μ , player A does not imi-

446 tate player B but instead redrew all strategy components from the same distributions
447 used to initialize the population. The mutation rate is set to zero in all simulations
448 except those presented in Supplementary Fig. S4 [SWP: *need to check*].

449 Unless otherwise specified, each simulation is run for 200,000 generations. When a
450 result is represented by a single summary value rather than as a temporal trajectory,
451 the metric was averaged over the final 10,000 generations.

452 To evaluate the dynamics of collective prediction, we define six summary met-
453 rics: collective accuracy, collective bias, collective variance, counterfactual collective
454 accuracy, interest diversity, and median reliance on AI. At each population state,
455 expectations and variances are evaluated analytically over the distributions of the
456 causal factors and AI errors rather than estimated from sampled prediction events.
457 Collective bias and collective variance constitute the bias–variance decomposition of
458 the group level collective prediction error.

459 Collective accuracy is normalized relative to the baseline prediction α_0 . A nega-
460 tive value indicates that the collective prediction performs worse than the baseline.
461 Counterfactual collective accuracy is evaluated by setting $\beta_A = 0$ while retaining the
462 evolved interests and beliefs of the AI-assisted population. In calculating interest
463 diversity, we use the convention that $\rho_i \ln \rho_i = 0$ when $\rho_i = 0$ (Table 1).

464 Data and code availability

465 All data are stored in a publicly available GitHub repository ([https://github.com/](https://github.com/parklab-snu/collective-intelligence-with-AI)
466 [parklab-snu/collective-intelligence-with-AI](https://github.com/parklab-snu/collective-intelligence-with-AI)).

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471 Contributions

472 Jaeseung Kim: Conceptualization, Formal Analysis, Investigation, Methodology,
473 Software, Visualization, Writing - Original Draft Preparation, Writing - Review &
474 Editing. Sang Woo Park: Conceptualization, Investigation, Methodology, Supervi-
475 sion, Writing - Original Draft Preparation, Writing - Review & Editing.

476 Competing Interests

477 The authors declare no competing interests.

Table 1: Evaluation metrics used to analyze collective prediction dynamics.

Metric	Definition	Interpretation
Collective accuracy	$1 - \frac{\mathbb{E}[(Y - \hat{Y}^{\text{re}})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$	Accuracy of the revised collective prediction relative to the baseline prediction α_0 .
Collective bias	$\left(\mathbb{E}[Y - \hat{Y}^{\text{re}}]\right)^2$	Squared expected deviation of the revised collective prediction from the true target.
Collective variance	$\text{Var}(Y - \hat{Y}^{\text{re}})$	Variability of the revised collective prediction error across prediction events.
Counterfactual collective accuracy	$1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$	Accuracy obtained after removing AI use from the current population while retaining its evolved interests and beliefs.
Interest diversity	$\exp\left(-\sum_{i=0}^M \rho_i \ln \rho_i\right)$	Effective number of interests represented in the population.
Median reliance on AI	$\text{median}_{A \in N}(\beta_A)$	Median reliance on AI in the population.

478 Supplementary Text

479 1 Omniscient AI

480 Omniscient AI assumes that AI can incorporate all causal factors regardless of the
481 player's interest.

482 1.1 Model

$$Y_A^{AI} = \alpha_0 + b_0 + \alpha_1 X_1 + b_1 + \cdots + \alpha_M X_M + b_M + \epsilon. \quad (S1)$$

483 For simplicity, prediction of Omniscient AI can be written as:

$$Y^{AI} = Y + B + \epsilon \quad (S2)$$

484 where Y stands for true prediction target ($Y = \alpha_0 + \alpha_1 X_1 + \cdots + \alpha_M X_M$) and
485 B is sum of all fixed biases ($B = b_0 + b_1 + \cdots + b_M$), ϵ follows normal distribution
486 with zero mean, τ standard deviation with no correlation within or between players.
487 Player A's prediction and revised prediction can be written as follows:

$$Y_A = c_A X_{e_A} \quad (S3)$$

$$Y_A^{re} = \beta_A Y^{AI} + (1 - \beta_A) Y_A \quad (S4)$$

489 where c_A stands for player A's belief on the factor, e_A stands for the interest, and
490 β_A stands for reliance on AI.

491 1.2 Aggregation

492 Because Y_A^{AI} is subject to all M factors, we use simple averaging aggregation for
493 collective prediction.

$$\begin{aligned} \hat{Y}^{re} &= \frac{1}{N} \sum_{j=1}^N Y_j^{re} \\ &= \frac{1}{N} \sum_{j=1}^N \{ \beta_j Y^{AI} + (1 - \beta_j) Y_j \} \\ &= \frac{1}{N} \sum_{j=1}^N \{ \beta_j (Y + B + \epsilon) + (1 - \beta_j) c_j X_{e_j} \} \end{aligned}$$

494 Reparameterization for simple derivation and interpretability.

$$\bar{\beta} := \frac{1}{N} \sum_{j=1}^N \beta_j \quad (S5)$$

495

$$\bar{\eta} := \frac{1}{N} \sum_{j=1}^N \beta_j \epsilon \quad (\text{S6})$$

496

$$\gamma_i := \frac{1}{N} \sum_{A \in \mathcal{A}_i} (1 - \beta_A) c_A \quad (\text{S7})$$

497 $\bar{\beta}$ can be interpreted as mean reliance on AI, γ stands for the coefficient for the factor
 498 i which is solely based on players' belief.

499 By this reparameterization, we can write $\hat{Y}^{re}$ as follows.

$$\hat{Y}^{re} = \bar{\beta}(Y + B) + \sum_{i=0}^M \gamma_i X_i + \bar{\eta} \quad (\text{S8})$$

500 1.3 Expected payoffs

501 Feedback

$$\delta_0 := \alpha_0 - (\bar{\beta}(\alpha_0 + B) + \gamma_0) \quad (\text{S9})$$

502

$$\delta_i := \alpha_i - (\bar{\beta}\alpha_i + \gamma_i), \quad (i \neq 0) \quad (\text{S10})$$

503

$$Y - \hat{Y}^{re} = \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta} \quad (\text{S11})$$

504 See equation Eq. (S6) for the $\bar{\eta}$.

505 δ_0 is constant term and δ_i is independent of α_k ($k \neq i$), ϵ , and $\bar{\eta}$.

$$\begin{aligned} \mathbb{E}(\pi_A) &= \mathbb{E}[Y_A^{re}(Y - \hat{Y}^{re})] \\ &= \mathbb{E} \left[\left\{ \beta_A \alpha_0 + \beta_A \sum_{i=1}^M \alpha_i X_i + \beta_A B + \beta_A \epsilon + (1 - \beta_A) c_A X_{e_A} \right\} \left\{ \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta} \right\} \right] \\ &= \beta_A \alpha_0 \delta_0 + \beta_A \sum_{i=1}^M \delta_i \alpha_i \sigma_i^2 + \beta_A B \delta_0 - \frac{1}{N} \beta_A^2 \tau^2 + (1 - \beta_A) c_A \delta_{e_A} \sigma_{e_A}^2 \\ &= \beta_A \left\{ (\alpha_0 + B) \delta_0 + \sum_{i=1}^M \alpha_i \delta_i \sigma_i^2 \right\} + (1 - \beta_A) c_A \delta_{e_A} \sigma_{e_A}^2 - \frac{\beta_A^2 \tau^2}{N} \end{aligned}$$

506 **Niche-expert**

507 If $e_A = 0$, player does not observe causal factor and their unrevised prediction is same
 508 with their belief c_A . Such that, the expected payoff is

$$\begin{aligned}
 \mathbb{E}(\pi_A) &= -\mathbb{E}[\rho_0(Y_A^{re} - Y)^2] \\
 &= -\mathbb{E}[\rho_0\{(\beta_A - 1)Y + \beta_A B + \beta_A \epsilon + (1 - \beta_A)c_A\}^2] \\
 &= -\mathbb{E}\left[\rho_0\left\{(\beta_A - 1)(\alpha_0 - c_A) + \beta_A B + (\beta_A - 1)\sum_{i=1}^M \alpha_i X_i + \beta_A \epsilon\right\}^2\right] \\
 &= -\rho_0\left\{(1 - \beta_A)^2(\alpha_0 - c_A)^2 - 2\beta_A(1 - \beta_A)(\alpha_0 - c_A)B + \beta_A^2 B^2\right. \\
 &\quad \left.+ (1 - \beta_A)^2 \sum_{i \neq 0} \alpha_i^2 \sigma_i^2 + \beta_A^2 \tau^2\right\}
 \end{aligned}$$

509 Else,

$$\begin{aligned}
 \mathbb{E}(\pi_A) &= -\mathbb{E}[\rho_{e_A}(Y_A^{re} - Y)^2] \\
 &= -\mathbb{E}[\rho_{e_A}\{(\beta_A - 1)Y + \beta_A B + \beta_A \epsilon + (1 - \beta_A)c_A X_{e_A}\}^2] \\
 &= -\mathbb{E}\left[\rho_{e_A}\left\{(\beta_A - 1)\alpha_0 + \beta_A B + (\beta_A - 1)(\alpha_{e_A} - c_A)X_{e_A} + (\beta_A - 1)\sum_{i \neq e_A, 0} \alpha_i X_i + \beta_A \epsilon\right\}^2\right] \\
 &= -\rho_{e_A}\left\{(1 - \beta_A)^2 \alpha_0^2 - 2\beta_A(1 - \beta_A)\alpha_0 B + \beta_A^2 B^2 + (1 - \beta_A)^2(\alpha_{e_A} - c_A)^2 \sigma_{e_A}^2\right. \\
 &\quad \left.+ (1 - \beta_A)^2 \sum_{i \neq e_A, 0} \alpha_i^2 \sigma_i^2 + \beta_A^2 \tau^2\right\}
 \end{aligned}$$

510 **Incentivizing or penalizing AI use**

511 Because additive term $\lambda\beta$ is not a random variable, we can simply derive the expected
 512 payoff as

$$\mathbb{E}(\pi_{A, Feedback}) = \mathbb{E}[Y_A^{re}(Y - \hat{Y}^{re}) + \lambda\beta_A] = \mathbb{E}[Y_A^{re}(Y - \hat{Y}^{re})] + \lambda\beta_A. \quad (\text{S12})$$

$$\mathbb{E}(\pi_{A, Niche-expert}) = -\mathbb{E}[\rho_{e_A}(Y_A^{re} - Y)^2 + \lambda\beta_A] = -\mathbb{E}[\rho_{e_A}(Y_A^{re} - Y)^2] + \lambda\beta_A. \quad (\text{S13})$$

513 **1.4 Expected accuracy**

$$\mathbb{E}[(Y - \alpha_0)^2] = \mathbb{E}\left[\left(\alpha_0 + \sum_{i=1}^M \alpha_i X_i - \alpha_0\right)^2\right] = \sum_{i=1}^M \alpha_i^2 \sigma_i^2 \quad (\text{S14})$$

514

$$\mathbb{E}[(Y - \hat{Y}^{re})^2] = \mathbb{E}\left[\left(\delta_0 + \sum_{i=1}^M \delta_i X_i - \hat{\eta}\right)^2\right] = \delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \frac{\tau^2}{N^2} \sum_{j=1}^N \beta_j^2 \quad (\text{S15})$$

515

$$\begin{aligned} R &= 1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]} \\ &= 1 - \frac{\delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \frac{\tau^2}{N^2} \sum_{j=1}^N \beta_j^2}{\sum_{i=1}^M \alpha_i^2 \sigma_i^2} \end{aligned}$$

516 2 Chatbot AI

517 2.1 Model

$$Y_A^{AI} = \alpha_{e_A} X_{e_A} + b_{e_A} + \epsilon \quad (\text{S16})$$

518 b_{e_A} stands for factor specific additive bias, ϵ follows normal distribution with zero
 519 mean, τ standard deviation with no correlation within or between players. Player
 520 A's prediction and revised prediction can be written as follows:

$$Y_A = c_A X_{e_A} \quad (\text{S17})$$

521

$$Y_A^{re} = \beta_A Y_A^{AI} + (1 - \beta_A) Y_A \quad (\text{S18})$$

522 where c_A stands for player A's belief on the factor, e_A stands for the interest, and
 523 β_A stands for reliance on AI.

524 2.2 Aggregation

525 For Chatbot AI, clustering aggregation is used for collective prediction.

$$\begin{aligned} \hat{Y}^{re} &= \sum_{i=0}^M \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} Y_A^{re} \\ &= \sum_{i=0}^M \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \{\beta_A Y_A^{AI} + (1 - \beta_A) Y_A\} \\ &= \sum_{i=0}^M \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} [\{\beta_A \alpha_{e_A} + (1 - \beta_A) c_A\} X_{e_A} + \beta_A b_{e_A} + \beta_A \epsilon] \end{aligned}$$

526 Reparameterize for the simplicity.

$$\tilde{c}_i := \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \{\beta_A \alpha_{e_A} + (1 - \beta_A) c_A\} \quad (\text{S19})$$

527

$$\tilde{B}_i := \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \beta_A b_{e_A} \quad (\text{S20})$$

528

$$\bar{B} := \sum_{i=0}^M \tilde{B}_i \quad (\text{S21})$$

529

$$\tilde{\eta}_i := \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \beta_A \epsilon \quad (\text{S22})$$

530

$$\bar{\eta} := \sum_{i=0}^M \tilde{\eta}_i \quad (\text{S23})$$

531 2.3 Expected payoffs

532 Feedback

533

$$\delta_0 := \alpha_0 - \tilde{c}_0 - \bar{B} \quad (\text{S24})$$

534

$$\delta_i := \alpha_i - \tilde{c}_i, \quad (i \neq 0) \quad (\text{S25})$$

$$Y - \hat{Y}^{re} = \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta} \quad (\text{S26})$$

535 δ_0 is constant term and δ_i is independent of α_k ($k \neq i$), ϵ , and $\bar{\eta}$.

$$\begin{aligned} \mathbb{E}(\pi_A) &= \mathbb{E}[Y_A^{re}(Y - \hat{Y}^{re})] \\ &= \mathbb{E} \left[\{(\beta_A \alpha_{e_A} + (1 - \beta_A) c_A) X_{e_A} + \beta_A b_{e_A} + \beta_A \epsilon\} \left\{ \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta} \right\} \right] \\ &= \{\beta_A \alpha_{e_A} + (1 - \beta_A) c_A\} \delta_{e_A} \sigma_{e_A}^2 + \beta_A b_{e_A} \delta_0 - \frac{1}{n_{e_A}} \beta_A^2 \tau^2 \end{aligned}$$

536 Niche-expert

537 If $e_A = 0$,

$$\begin{aligned} \mathbb{E}(\pi_A) &= -\mathbb{E}[\rho_0(Y_A^{re} - Y)^2] \\ &= -\mathbb{E} \left[\rho_0 \left[\{ \beta_A \alpha_0 + (1 - \beta_A) c_A \} + \beta_A b_0 + \beta_A \epsilon - \alpha_0 - \sum_{i=1}^M \alpha_i X_i \right]^2 \right] \\ &= -\rho_0 \left\{ (1 - \beta_A)^2 (c_A - \alpha_0)^2 + 2(1 - \beta_A)(c_A - \alpha_0) \beta_A b_0 + \beta_A^2 b_0^2 \right. \\ &\quad \left. + \sum_{i \neq 0} \alpha_i^2 \sigma_i^2 + \beta_A^2 \tau^2 \right\} \end{aligned}$$

538 Else,

$$\begin{aligned}
\mathbb{E}(\pi_A) &= -\mathbb{E}[\rho_{e_A}(Y_A^{\text{re}} - Y)^2] \\
&= -\mathbb{E}\left[\rho_{e_A}\left[\{\beta_A\alpha_{e_A} + (1 - \beta_A)c_A\}X_{e_A} + \beta_A b_{e_A} + \beta_A\epsilon - \alpha_0 - \sum_{i=1}^M \alpha_i X_i\right]^2\right] \\
&= -\rho_{e_A}\left\{(1 - \beta_A)^2(c_A - \alpha_{e_A})^2\sigma_{e_A}^2 + \sum_{i \neq e_A, 0} \alpha_i^2 \sigma_i^2\right. \\
&\quad \left.+ \alpha_0^2 + \beta_A^2 b_{e_A}^2 - 2\alpha_0\beta_A b_{e_A} + \beta_A^2 \tau^2\right\}
\end{aligned}$$

539 Incentivizing or penalizing AI use

540 Because additive term $\lambda\beta$ is not a random variable, we can simply derive the expected
541 payoff as

$$\mathbb{E}(\pi_{A, \text{Feedback}}) = \mathbb{E}\left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) + \lambda\beta_A\right] = \mathbb{E}\left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}})\right] + \lambda\beta_A. \quad (\text{S27})$$

$$\mathbb{E}(\pi_{A, \text{Niche-expert}}) = -\mathbb{E}\left[\rho_{e_A}(Y_A^{\text{re}} - Y)^2 + \lambda\beta_A\right] = -\mathbb{E}\left[\rho_{e_A}(Y_A^{\text{re}} - Y)^2\right] + \lambda\beta_A. \quad (\text{S28})$$

542 Balanced

543 Expected payoff of the balanced structure can also be driven by arithmetic mean of
544 expected payoff of feedback and niche-expert.

$$\mathbb{E}(\pi_A) = \mathbb{E}\left[\frac{Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) - \rho_{e_A}(Y_A^{\text{re}} - Y)^2}{2}\right] = \frac{\mathbb{E}\left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}})\right] - \mathbb{E}\left[\rho_{e_A}(Y_A^{\text{re}} - Y)^2\right]}{2} \quad (\text{S29})$$

545 2.4 Expected accuracy

$$\mathbb{E}[(Y - \alpha_0)^2] = \mathbb{E}\left[\left(\alpha_0 + \sum_{i=1}^M \alpha_i X_i - \alpha_0\right)^2\right] = \sum_{i=1}^M \alpha_i^2 \sigma_i^2 \quad (\text{S30})$$

$$\mathbb{E}[(Y - \hat{Y})^2] = \mathbb{E}\left[\left(\delta_0 + \sum_{i=1}^M \delta_i X_i - \hat{\eta}\right)^2\right] = \delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \sum_{i=0}^M \frac{\tau^2}{n_i^2} \sum_{A \in \mathcal{A}_i} \beta_A^2 \quad (\text{S31})$$

$$\begin{aligned}
R &= 1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]} \\
&= 1 - \frac{\delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \sum_{i=0}^M \frac{\tau^2}{n_i^2} \sum_{A \in \mathcal{A}_i} \beta_A^2}{\sum_{i=1}^M \alpha_i^2 \sigma_i^2}
\end{aligned}$$

548 3 Evolutionary dynamics

549 For sufficiently large population ($N \rightarrow \infty$), we can approximate the deterministic
550 evolutionary dynamics using the replicator equation. Let λ denote a strategy, p_λ its
551 frequency, and $u_\lambda = \mathbb{E}(\pi_\lambda)$ its expected payoff. The replicator equation is

$$\frac{dp_\lambda}{dt} = p_\lambda(u_\lambda - \bar{u}), \quad (\text{S32})$$

552 where $\bar{u} = \sum_\lambda p_\lambda u_\lambda$. Therefore, at an internal stationary state, all strategies
553 with positive frequency have the same expected payoff. A boundary state can also
554 be stationary because a strategy with exactly zero frequency cannot increase under
555 the replicator equation. Thus, stability of a boundary state needs to be considered
556 separately by examining invasion by a rare strategy.

557 3.1 Omniscient AI

558 3.1.1 Feedback

559 When the population size approaches infinity, the effect of stochastic error on the
560 collective prediction vanishes ($\bar{\eta} \rightarrow 0$) and the effect of stochastic error in the ex-
561 pected payoff, $-\beta^2 \tau^2 / N$, also vanishes. Following prior research (Wang et al., 2025),
562 we define the Lyapunov function as the expected error of collective prediction

$$V = \mathbb{E} \left[(Y - \hat{Y}^{re})^2 \right]. \quad (\text{S33})$$

563 The function V is non-negative and equals zero when the collective prediction is
564 identical to the target. To examine how V changes under the evolutionary dynamics,
565 we take its time derivative:

$$\begin{aligned}
\frac{dV}{dt} &= -2\mathbb{E} \left[(Y - \hat{Y}^{re}) \frac{d\hat{Y}^{re}}{dt} \right] \\
&= -2\mathbb{E} \left[\left(\delta_0 + \sum_{i=1}^M \delta_i X_i \right) \left\{ \frac{d\bar{\beta}}{dt} (Y + B) + \sum_{i=0}^M \frac{d\gamma_i}{dt} X_i \right\} \right] \\
&= -2 \left[\left\{ (\alpha_0 + B) \delta_0 + \sum_{i=1}^M \alpha_i \delta_i \sigma_i^2 \right\} \frac{d\bar{\beta}}{dt} + \delta_0 \frac{d\gamma_0}{dt} + \sum_{i=1}^M \delta_i \sigma_i^2 \frac{d\gamma_i}{dt} \right].
\end{aligned}$$

566 For notational convenience, we define

$$K = (\alpha_0 + B) \delta_0 + \sum_{i=1}^M \alpha_i \delta_i \sigma_i^2. \quad (\text{S34})$$

567 Although K changes with the population state, it is common to all strategies at a
568 given state and can therefore be treated as a constant with respect to the strategy
569 index λ . We also define

$$G_e = \begin{cases} \delta_0, & e = 0, \\ \delta_e \sigma_e^2, & e \neq 0. \end{cases} \quad (\text{S35})$$

570 By the definitions of $\bar{\beta}$ and γ_i and the replicator equation,

$$\frac{d\bar{\beta}}{dt} = \sum_{\lambda} p_{\lambda} (u_{\lambda} - \bar{u}) \beta_{\lambda} = \text{Cov}(\beta, u). \quad (\text{S36})$$

571 Similarly, the terms involving γ_i can be combined as

$$\delta_0 \frac{d\gamma_0}{dt} + \sum_{i=1}^M \delta_i \sigma_i^2 \frac{d\gamma_i}{dt} = \text{Cov}((1 - \beta) c G_e, u). \quad (\text{S37})$$

572 Therefore,

$$\begin{aligned}
\frac{dV}{dt} &= -2 \{ K \text{Cov}(\beta, u) + \text{Cov}((1 - \beta) c G_e, u) \} \\
&= -2 \text{Cov}(K\beta + (1 - \beta) c G_e, u).
\end{aligned}$$

573 For the feedback structure in the limit $N \rightarrow \infty$, the expected payoff of strategy
574 λ is

$$u_{\lambda} = K \beta_{\lambda} + (1 - \beta_{\lambda}) c_{\lambda} G_{e_{\lambda}}. \quad (\text{S38})$$

575 Which is same with Thus,

$$\frac{dV}{dt} = -2 \text{Cov}(u, u) = -2 \text{Var}(u) \leq 0. \quad (\text{S39})$$

Therefore, the expected error of collective prediction cannot increase during the evolutionary process and decreases whenever strategies differ in their expected payoffs. The derivative becomes zero when the variance of expected payoffs is zero, $\text{Var}(u) = 0$. This condition corresponds to equal expected payoffs among the strategies present in the population. A boundary state in which the entire population consists of a single strategy can also satisfy $\text{Var}(u) = 0$ even when $V > 0$. However, such a state is not evolutionarily stable under the feedback structure. If the collective prediction differs from the target Y , a nonzero residual remains. By the definition of the feedback payoff, a novel strategy whose prediction aligns with the direction of this residual obtains a higher expected payoff than the resident strategy and can therefore invade the population. Thus, an imperfect boundary equilibrium with $V > 0$ is unstable to the introduction of novel predictions.

Consequently, the evolutionary dynamics continue to reduce the collective prediction error until the population reaches the stable state with zero residual, $V = 0$. At this state, the collective prediction equals the target, $\hat{Y}^{re} = Y$, and the collective accuracy reaches $R = 1$.

3.1.2 Niche-expert

Unlike the stochastic error in the collective prediction, stochastic error in the niche-expert payoff is evaluated at the individual level and does not vanish as population size increases. Therefore, we retain $\tau > 0$ in the following derivation.

We first consider strategies having the same interest by optimizing the expected payoff with respect to c and β . For a nonzero interest e , the expected payoff is strictly concave with respect to c for $\beta < 1$ and $\sigma_e > 0$, and is maximized at

$$c_e^* = \alpha_e. \quad (\text{S40})$$

After substituting $c_e^* = \alpha_e$, the payoff is strictly concave with respect to β . Its unconstrained stationary point is

$$\tilde{\beta}_e = \frac{\alpha_0^2 + \alpha_0 B + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2}{(\alpha_0 + B)^2 + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S41})$$

Because β is constrained between 0 and 1, the optimal β_e^* is $\tilde{\beta}_e$ when $0 < \tilde{\beta}_e < 1$, and otherwise lies at the corresponding boundary, $\beta_e^* = 0$ or $\beta_e^* = 1$. Thus, the boundary cases do not require a separate evolutionary argument; they only change the optimal prediction and its minimum expected squared error. When $\beta_e^* = 1$, the revised prediction is independent of c_e , so different values of c_e are selectively neutral and can be treated as a single payoff-equivalent strategy class.

Let L_e denote the minimum expected individual error, $\mathbb{E}[(Y_A^{re} - Y)^2]$, within interest e . For an internal optimum,

$$L_e = \frac{B^2 \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 + \tau^2 \left(\alpha_0^2 + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 \right)}{(\alpha_0 + B)^2 + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S42})$$

609 If the optimum is located at $\beta_e^* = 0$ or $\beta_e^* = 1$, L_e is instead obtained by evaluating
 610 the expected squared prediction error at the corresponding boundary.

611 For interest 0, optimization with respect to c gives

$$c_0^*(\beta) = \alpha_0 - \frac{\beta}{1 - \beta} B. \quad (\text{S43})$$

612 Substituting this into the expected payoff gives a strictly concave function of β ,
 613 with the unique optimum

$$\beta_0^* = \frac{\sum_{j=1}^M \alpha_j^2 \sigma_j^2}{\sum_{j=1}^M \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S44})$$

614 Therefore,

$$c_0^* = \alpha_0 - \frac{B}{\tau^2} \sum_{j=1}^M \alpha_j^2 \sigma_j^2, \quad (\text{S45})$$

615 and

$$L_0 = \frac{\tau^2 \sum_{j=1}^M \alpha_j^2 \sigma_j^2}{\sum_{j=1}^M \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S46})$$

616 We denote the optimal strategy, or the payoff-equivalent strategy class when
 617 $\beta_e^* = 1$, by S_e . A state containing a suboptimal strategy within an interest can
 618 be stationary if the optimal strategy is absent, but it is unstable against invasion
 619 because the optimal strategy has a strictly higher payoff at the same ρ_e . Therefore,
 620 a stable state can contain only S_e within each represented interest.

621 We next consider the frequencies of the $M + 1$ optimal strategies. Let ρ_e denote
 622 the frequency of S_e . Its expected payoff is

$$u_e = -\rho_e L_e. \quad (\text{S47})$$

623 Assuming $L_e > 0$ for all interests, the equal-payoff condition of the replicator
 624 equation at an internal stationary state gives

$$\rho_0^* L_0 = \rho_1^* L_1 = \cdots = \rho_M^* L_M. \quad (\text{S48})$$

625 Together with $\sum_{e=0}^M \rho_e^* = 1$, this uniquely determines

$$\rho_e^* = \frac{D}{L_e}, \quad D = \left(\sum_{j=0}^M \frac{1}{L_j} \right)^{-1}. \quad (\text{S49})$$

626 This internal stationary state is locally stable. At equilibrium, all represented
 627 interests have the same expected payoff, with $\rho_e^* L_e = D$. If the frequency of one
 628 interest increases slightly above its equilibrium value, its payoff decreases because
 629 the niche-expert payoff is proportional to $-\rho_e L_e$. Conversely, if the frequency of
 630 an interest decreases below its equilibrium value, its payoff increases. The replicator
 631 dynamics therefore generate a restoring force that reduces small deviations in interest
 632 frequencies and drives the population back toward the internal stationary state.

633 If one or more interests are absent, the resulting boundary state can be stationary
 634 under the replicator equation, but it is unstable against invasion. The payoff of a rare
 635 absent interest approaches zero as $\rho_e \rightarrow 0$, whereas the mean payoff of the resident
 636 population is negative when $L_e > 0$, so the rare interest increases when introduced
 637 at sufficiently small frequency.

638 Thus, there exists a unique stable full-support equilibrium in terms of the optimal
 639 prediction strategies and their frequencies. Whether the constrained optimum has
 640 $\beta_e^* = 0$, $0 < \beta_e^* < 1$, or $\beta_e^* = 1$ does not change the frequency argument, because
 641 these cases differ only through the resulting value of L_e . When $\beta_e^* = 1$, the strategy
 642 is selectively neutral with respect to c_e , but its prediction, payoff, and equilibrium
 643 interest frequency remain uniquely determined.

644 This equilibrium also allows a direct comparison with the expected accuracy of
 645 the AI. Using Y^{AI} itself without aggregation as the prediction gives

$$\mathbb{E} [(Y - Y^{\text{AI}})^2] = B^2 + \tau^2. \quad (\text{S50})$$

646 For every interest, complete reliance on AI, $\beta = 1$, is a feasible strategy. There-
 647 fore, because L_e is the minimum individual expected squared prediction error within
 648 interest e ,

$$L_e \leq B^2 + \tau^2. \quad (\text{S51})$$

649 Moreover, for interest 0,

$$L_0 = \frac{\tau^2 \sum_{j=1}^M \alpha_j^2 \sigma_j^2}{\sum_{j=1}^M \alpha_j^2 \sigma_j^2 + \tau^2} < \tau^2 \leq B^2 + \tau^2. \quad (\text{S52})$$

650 Because the stable equilibrium has $\rho_0^* > 0$,

$$\sum_{e=0}^M \rho_e^* L_e < B^2 + \tau^2. \quad (\text{S53})$$

651 The collective prediction is the average of individual revised predictions. By
 652 convexity of the squared error,

$$\mathbb{E} [(Y - \hat{Y}^{re})^2] \leq \sum_{e=0}^M \rho_e^* L_e. \quad (\text{S54})$$

Therefore,

$$\mathbb{E}[(Y - \hat{Y}^{re})^2] < \mathbb{E}[(Y - Y^{AI})^2]. \quad (\text{S55})$$

Because the collective accuracy and AI accuracy use the same normalization by $\mathbb{E}[(Y - \alpha_0)^2]$, the stationary collective accuracy under the niche-expert structure is strictly greater than the expected accuracy of the AI prediction.

Although the niche-expert structure improves collective prediction relative to the AI, the stationary collective accuracy does not reach one when $\tau > 0$. For a nonzero factor i , the residual coefficient of the stationary collective prediction can be written as

$$\delta_i^* = \alpha_i \{1 - \bar{\beta}^* - \rho_i^*(1 - \beta_i^*)\} = \alpha_i \sum_{e \neq i} \rho_e^*(1 - \beta_e^*). \quad (\text{S56})$$

At the stable equilibrium, $\rho_0^* > 0$. In addition, $\tau > 0$ implies $\beta_0^* < 1$. Therefore, for every factor with $\alpha_i \neq 0$,

$$\delta_i^* \neq 0. \quad (\text{S57})$$

Because $\sum_{i=1}^M \alpha_i^2 \sigma_i^2 > 0$, at least one factor contributes positive prediction error. Hence,

$$\mathbb{E}[(Y - \hat{Y}^{re})^2] > 0, \quad (\text{S58})$$

and consequently the stationary collective accuracy satisfies $R^* < 1$.

3.2 Chatbot AI

3.2.1 Feedback

With Chatbot AI, the effect of stochastic error on the collective prediction depends on n_i , requiring the additional assumption that $n_i \rightarrow \infty$ for every interest for the stochastic error to vanish. However, Under this assumption, the loss of interest diversity cannot occur, and therefore the stochastic breakdown of collective accuracy observed in finite populations cannot be analytically captured. Instead, by assuming that each interest is represented by infinitely many players, we can show that the expected collective prediction error monotonically decreases until perfect accuracy is achieved. Because finite population dynamics are more relevant to the empirical populations considered in this study, we retain finite population simulations for the main result.

3.2.2 Niche-expert

We first consider strategies having the same interest by optimizing the expected payoff with respect to c and β . For a nonzero interest e , the expected payoff is strictly concave with respect to c for $\beta < 1$ and $\sigma_e > 0$, and is maximized at

$$c_e^* = \alpha_e. \quad (\text{S59})$$

682 After substituting $c_e^* = \alpha_e$, the payoff is strictly concave with respect to β . Its
 683 unconstrained stationary point is

$$\tilde{\beta}_e = \frac{2\alpha_0 b_e}{2b_e^2 + 2\tau^2}. \quad (\text{S60})$$

684 Because β is constrained between 0 and 1, the optimal β_e^* is $\tilde{\beta}_e$ when $0 < \tilde{\beta}_e < 1$,
 685 and otherwise lies at the corresponding boundary, $\beta_e^* = 0$ or $\beta_e^* = 1$. Because we
 686 assumes AI model with moderate accuracy, b_e and τ is relatively smaller than α_0
 687 which makes $\tilde{\beta}_e > 1$ when $\alpha_0 b_e > 0$ and $\tilde{\beta}_e < 0$ when $\alpha_0 b_e < 0$. These boundary
 688 cases lead to extreme reliance on AI or complete exclusion of AI which both hinders
 689 the emergence of CI. Only when the prediction task is centered near zero and AI is
 690 relatively inaccurate ($\alpha_0 b_e < b_e^2 + \tau^2$), and $\alpha_0 b_e > 0$, the internal optimum β_e^* can be
 691 exist.

692 For interest 0, optimization with respect to c gives

$$c_0^*(\beta) = \alpha_0 - \frac{\beta}{1 - \beta} b_0. \quad (\text{S61})$$

693 Substituting this into the expected payoff gives a strictly concave function of β , with
 694 the unique optimum

$$\beta_0^* = 0. \quad (\text{S62})$$

Supplementary Figure

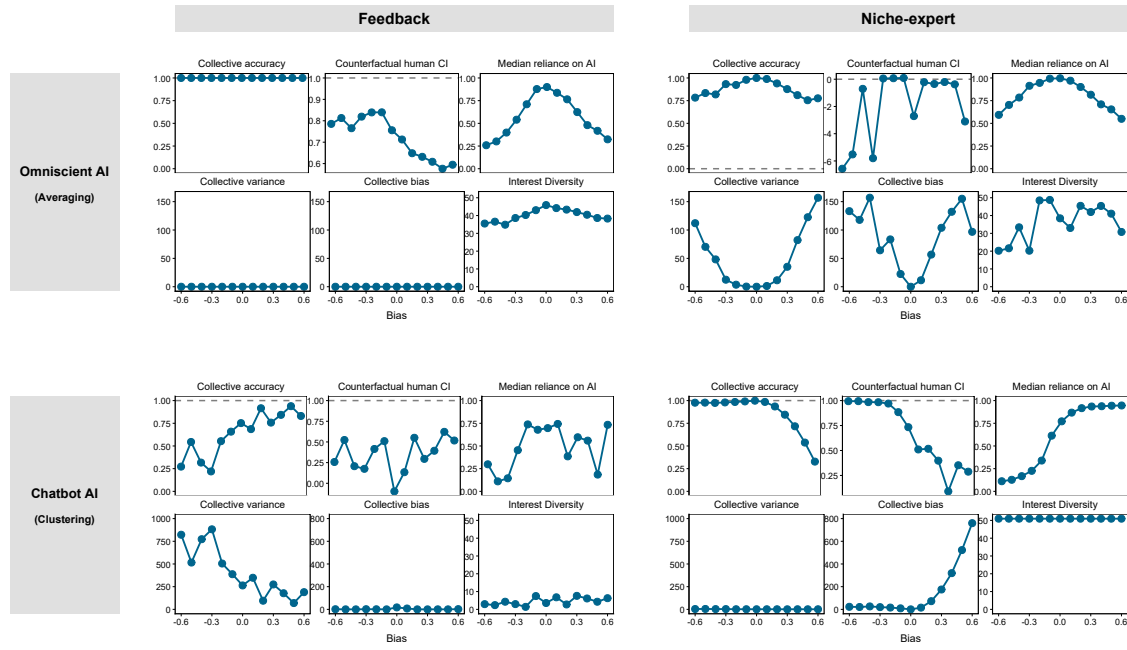


Figure S1: Caption. Text.

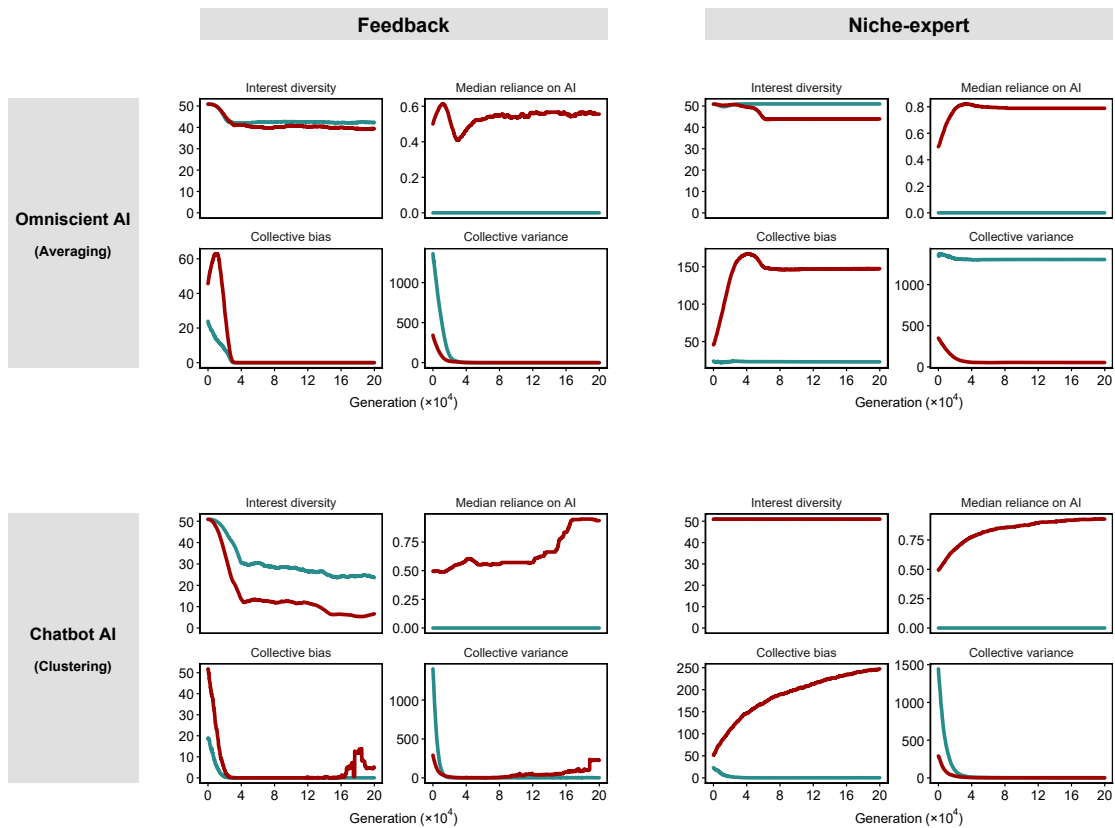


Figure S2: **Caption.** Text.

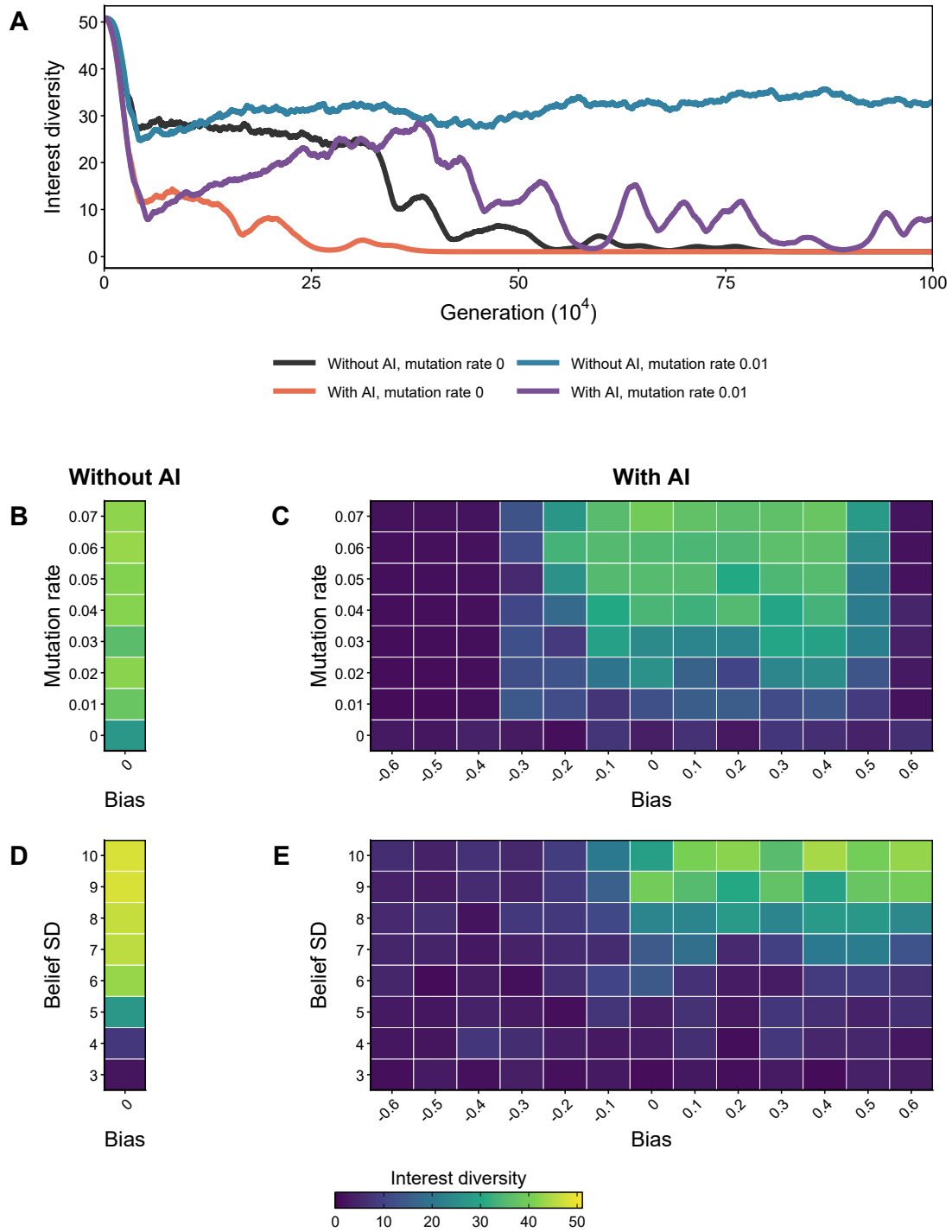


Figure S3: **Caption.** Text.

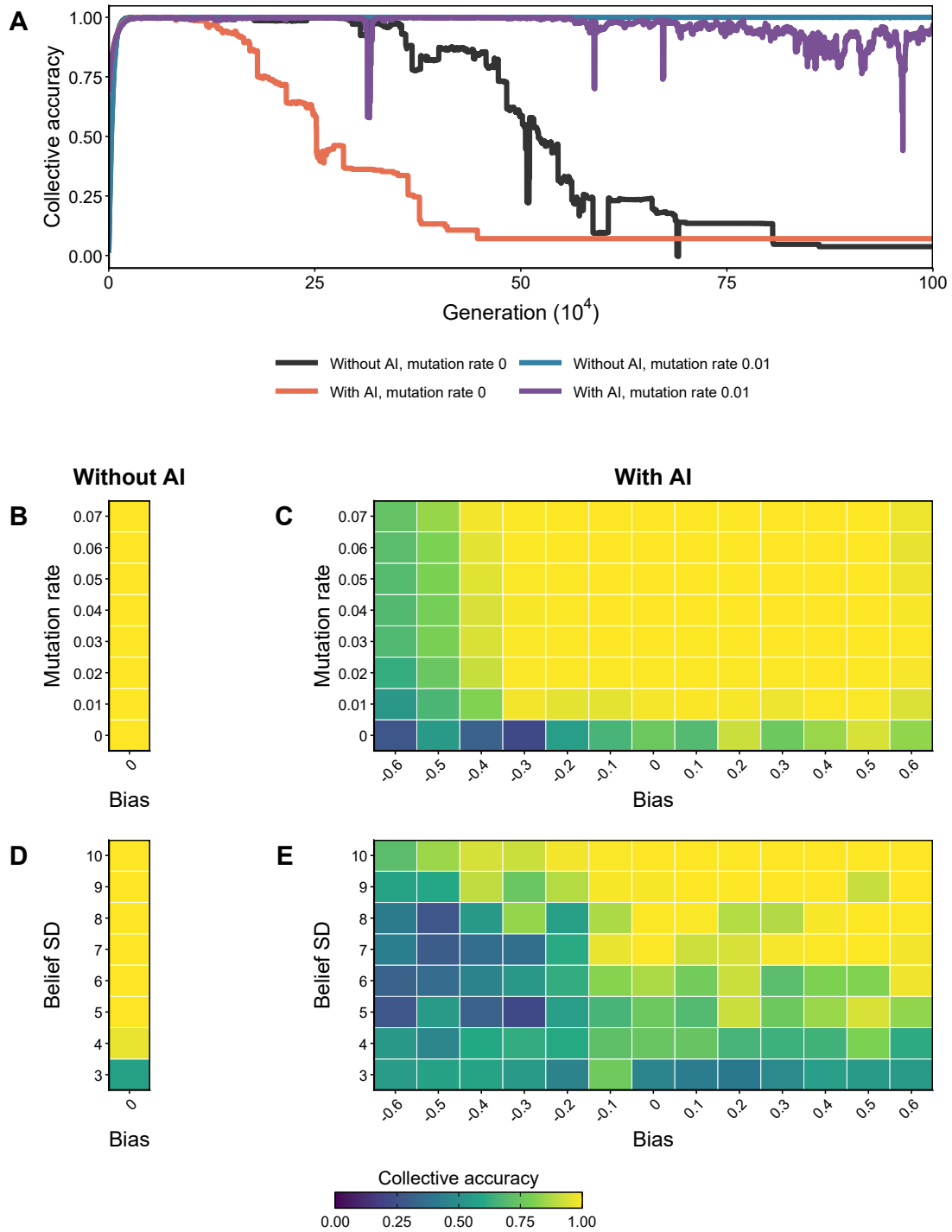


Figure S4: **Caption.** Text.

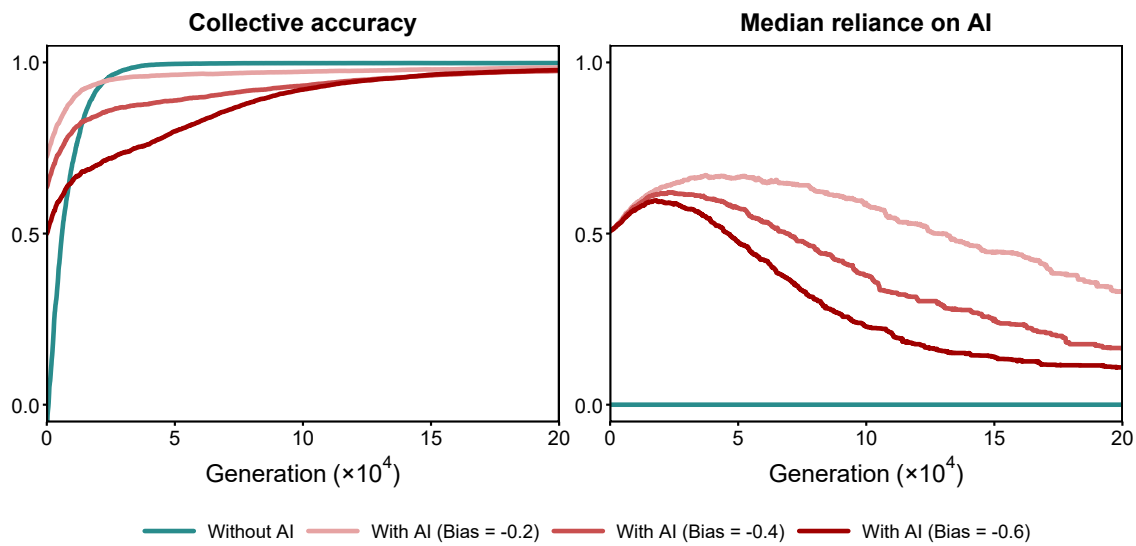


Figure S5: **Caption.** Text.

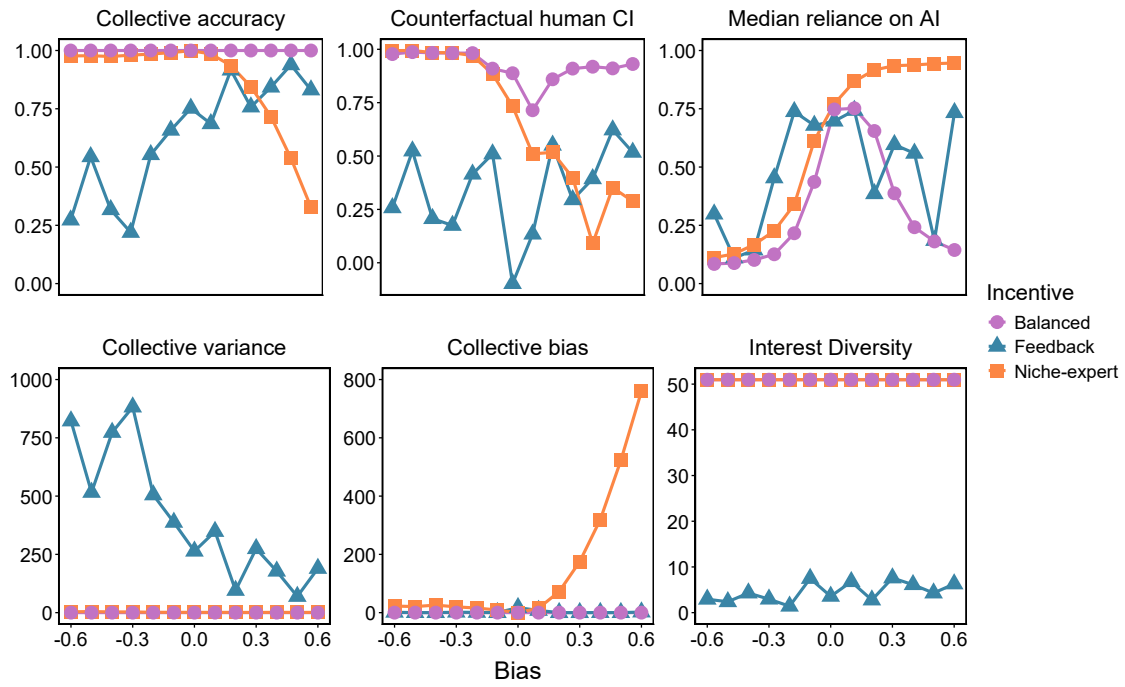


Figure S6: **Caption.** Text.

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